Healthcare Data Science

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Welcome

This is a work-in-progress draft. Chapters, structure, and content are still evolving — feedback is welcome, but please don't cite or redistribute this version.

> Note
>
> **Status:** Early draft, actively being written.

Preface

About the Author

Omar Soliman is a Lead Data Scientist at a Fortune 5 healthcare company, where he has worked for over 8 years. He is also an adjunct faculty member in Data Science at Massachusetts College of Pharmacy and Health Sciences (MCPHS),[1] where he teaches MBA students how to use data science for healthcare decision-making and helps medical directors and executives make data-driven decisions that improve patient outcomes. Omar earned his Master of Medical Sciences in Clinical Investigation (Clinical Data Science track) from Harvard Medical School in 2018, completing his thesis at the Dana-Farber Cancer Institute. He is passionate about simplifying complex data science concepts and teaching aspiring data scientists how to apply their skills in real-world healthcare scenarios. In his free time, Omar enjoys walking and running outdoors, watching documentaries, spending time with his two kids, traveling, and learning about the latest advances in AI in healthcare.

Motivation for the Book

While preparing for his courses at MCPHS, Omar realized there was a lack of comprehensive, up-to-date resources focused on the practical application of data science in healthcare. Existing resources were often too theoretical,

[1] I had to autocomplete "Massachusetts" despite living here for over 10 years. :)

too general (not healthcare-specific), or outdated, leaving students and professionals without a clear path to apply their knowledge effectively. This book was born out of a desire to fill that gap, providing practical, case-based examples that help readers learn how to apply data science in healthcare settings.

This Book Is For

- New, aspiring, and junior data scientists who want an organized and practical way to learn healthcare data science.
- MBA students, project managers, and healthcare professionals who want to understand how data science can be applied to improve healthcare decision-making.
- Experienced data scientists from other fields who want to transition into healthcare and need a practical guide to get started.

How this book is organized

This book is structured around the three main tasks of data science, a framework I learned from my mentor Miguel Hernán, Professor of Epidemiology and Biostatistics at Harvard: **description**, **prediction**, and **causal inference**. Each task gets its own section, with practical examples and case studies showing how to apply these techniques in healthcare settings. Earlier chapters ground you in the lifecycle of healthcare data — where it comes from and where it lives — before we get to the analysis.

Along the way, you'll be introduced to three programming languages — SQL, Python, and R — and to visualization tools like Tableau. One of my main goals is to take the anxiety out of learning to program. To that end, we lean on browser-based environments like Google Colab and Posit Cloud (formerly RStudio Cloud), so you can start writing and running code immediately, with no complex installation. We'll also show you how to use AI to assist your coding.

All the code in this book is available on GitHub, and I encourage you to explore it, run it, and adapt it to your own needs. Now — relax, and enjoy learning healthcare data science at your own pace.

1 Healthcare Data

In this chapter, we'll explore the most common sources of healthcare data. Most of these are universal — they'll look familiar wherever in the world you practice. The exception is insurance claims data, which is heavily shaped by the US system. If you're reading from outside the US and that section feels unintuitive, that's expected: it's a quirk of how American healthcare is financed, not a gap in your understanding.

The sources we'll cover:

1. Insurance claims (administrative claims data)
2. Electronic health records (EHRs)
3. Other sources

1.1 Insurance claims

Before diving into claims data, let's take a few minutes on health insurance in the US — a refresher for some, a first pass for others.

> **Note**
>
> When we say *insurance* in this book, we mean **health insurance** — not auto, home, or any other kind.

At some point, having health insurance in the US was effectively mandatory: you could face a penalty for going without it. As of 2026, whether that's still true depends on which state you live in.[1] And because care is so expensive here, paying out of pocket isn't realistic for most people — which is exactly why claims data exists in the volume and shape that it does.

So what actually happens? When you see a provider — a doctor, dentist, nurse, and so on — they diagnose you, perform procedures, and/or prescribe medication. To get paid, they submit a record of all of this to the insurer. That record is the **claim**: they're *claiming* payment for the service. This is why insurance claims data is geared mainly toward billing, not clinical detail.

But there are a *ton* of diseases, procedures, and drugs out there. Do providers just submit free text? Good question — no. They submit **codes**: diagnosis codes, procedure codes, and drug codes. Let's take each in turn: what it covers, who maintains it, and why it matters for your analysis.

1.1.1 Diagnosis codes (ICD codes)

The official name for diagnosis codes is International Classification of Diseases (ICD) codes, maintained by the World Health Organization (WHO). The latest version as of 2026 is ICD-11. Most countries use the WHO's ICD codes directly, but some use a country-specific modification instead. The US, for example, uses the Clinical Modification, known as ICD-10-CM. Australia, New Zealand, Ireland, and Saudi Arabia use the Australian Modification, ICD-10-AM. ICD codes are alphanumeric — they consist of both letters and numbers. For example, I10 is the ICD-10 code for primary/essential hypertension.

This matters for your analysis: code sets change over time. The US switched from ICD-9 to ICD-10 in 2015, so if your dataset spans that switch — say, some records from 2014 and some from 2017 — you'll notice

[1] Massachusetts got there first, requiring residents to carry coverage in 2006 — a few years ahead of the federal version. So if you're reading from my home state, you've been mandated longer than most.

the codes for the same condition look completely different in each, and you can't just combine the two as-is; you need to translate one set into the other first, or your trends over time will be wrong.

1.1.2 Procedure codes (CPT codes)

The official name for procedure codes is Current Procedural Terminology (CPT) codes, maintained by the American Medical Association (AMA). They're used mostly in the US, but the AMA licenses CPT to more than 60 other countries as well. CPT codes are five characters long, either all numeric or a mix of digits and letters depending on the category. I'll avoid going too deep into medical coding systems here, so the book stays focused on healthcare data science — just enough to do your job. An example is 99213, the code for an established-patient (returning, not new) office or outpatient evaluation and management visit. When insurance companies and healthcare providers negotiate prices, they do so based on these CPT codes rather than going back and forth in free text, which would be impractical at today's scale.

1.1.3 Drug codes (NDC codes)

The official name for drug codes is National Drug Code (NDC), maintained by the Food and Drug Administration (FDA). Every drug marketed in the US — prescription or over-the-counter — gets an NDC code once its manufacturer lists it with the FDA. Note that having an NDC code doesn't by itself mean the drug is FDA-approved: OTC drugs marketed under an FDA monograph get one too, without going through full approval. Next time you're at a pharmacy, check an over-the-counter medicine — pain relievers, for example — and you'll see the NDC code printed right on the package.

One gotcha: NDCs are more specific than just "the drug" — they also encode the manufacturer and package size, so the exact same drug and dose can have several different NDCs, and old ones can be retired and replaced when a product is repackaged. If your question is "which drug," not "which package," you'll usually want to group NDCs up to the ingredient level rather than using the raw codes as-is.

1.1.4 Other codes

- **GPI** (Generic Product Identifier) — a drug classification (from Wolters Kluwer) that groups drugs by therapeutic class rather than by exact package, useful when NDC is more detail than you need.
- **DRG** (Diagnosis-Related Group) — bundles an entire hospital stay into one billing category, used to pay hospitals for inpatient care.
- **HCPCS** (Healthcare Common Procedure Coding System) — CPT codes are actually "HCPCS Level I"; Level II covers things like medical supplies, equipment, and ambulance rides.
- **LOINC** (Logical Observation Identifiers Names and Codes) — standard codes for lab tests and other clinical measurements.
- **SNOMED-CT** (Systematized Nomenclature of Medicine – Clinical Terms) — a more detailed clinical vocabulary than ICD, used inside EHRs to record diagnoses, symptoms, and findings at the point of care.

1.2 Electronic health records (EHR)

When you visit a healthcare professional — a doctor, for example (in the US, generically called a "provider") — you might see them typing on their computer. They're documenting your visit. Sometimes you might notice another person in the room, called a scribe, whose job is to help the provider take notes. In recent years, there's been a huge influx of AI tools called AI scribes that listen to the conversation and produce summaries.

So what's in the EHR? It contains demographic data about the patient — name, age, ID, gender, address, etc. It also contains medications, lab results, immunizations, visit notes, vital signs, and more. The largest EHR vendors are Epic and Oracle Health (formerly Cerner), among others.

Data in EHRs can be structured or unstructured (free text) — the same information, recorded two different ways:

	Structured	Unstructured (free text)
Glucose reading	Glucose: 7.2 mmol/L	"Patient's sugar looked a bit high today, will recheck."
Hypertension	I10 (diagnosis code)	"History of high blood pressure, on medication."

Structured fields are ready to pull into a spreadsheet or a model as-is. Unstructured notes hold plenty of clinically useful detail too, but getting it out takes more work than just reading a field.

1.3 Data storage

1.4 How to interact with healthcare data? (SQL)

2 Description

(Placeholder — content coming soon.)

3 Prediction

(Placeholder — content coming soon.)

4 Causal Inference

This chapter is about **causal** inference — not *casual* inference, though I promise that mix-up trips people up more often than you'd think.

My sister wanted to buy a laptop and was uncertain whether a Windows or Mac is better. My 8-year-old son was listening in, and he said, "I recommend Windows." When I asked why, he replied that it's because his dad has a Mac and his mom has Windows, and when he checked for information leaks online, he found more of his dad's information exposed. From that, my son concluded that owning a MacBook makes your information publicly available. We'll revisit this story toward the end of the chapter. But I wanted to emphasize that we start making causal inference claims this early in life — despite that fancy name, "causal inference."

In September 2025, the US administration publicly claimed that acetaminophen (Tylenol) use in pregnancy causes autism (a neurodevelopmental condition) in children. My goal here is to walk you through how to think critically about a claim like this — verifying it and scientifically critiquing it, rather than taking a side on it.

Last example before diving right in. We frequently make causal claims in our life — for example, if your sibling develops a stomachache after eating at a restaurant you didn't eat at, and you were fine, you'd likely conclude that the meal is what caused the stomachache, especially since you both live in the same house and do most activities together. We'll come back to this exact story later in this chapter.